

Intro to Data Science

Lab 4

Optimization with Linear Programming

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1 What-if Analysis in Excel

What-if analysis is a commonly used data analysis technique that allows you to test various scenarios and determine a range of possible outcomes. In other words, it lets you see the impact of making a specific change without changing the real data. Such analysis is helpful in financial modeling, forecasting, and other business applications where backward calculations are required.

Goal Seek is an Excel What-If Analysis tool that allows users to find the necessary input value in a formula to achieve a specific target output. While other What-If Analysis tools, like Data Tables or Scenario Manager, consider multiple variables, Goal Seek focuses on adjusting a single variable to meet a desired result.

File to use: GoalSeek-Examples

The Initial worksheet shows the various examples and the Solution worksheet shows the solution with the required variable value.

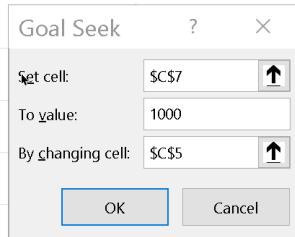
To use GoalSeek effectively, you need to set up the data so that you have a formula cell and a variable cell whose value when changed will affect the formula cell value.

Goal Seek Example #1: The Revenue calculation is given in C7 (formula cell). The problem is to determine how many items to sell in C5 (variable cell) in order to achieve a certain revenue, for e.g. 1000

Go to Data -> What if Analysis -> Goal Seek.

In the Goal Seek dialog box, define the cells/values to test

- Set cell - the reference to the formula cell (C7).
- To value - the formula result you are trying to achieve (1000).
- By changing cell - the reference for the input cell that you want to adjust (C5).



and click OK:

After Goal Seek is complete, you can perform an undo (Ctrl – Z) to restore the original input values so that you can repeat this for a different scenario.

For e.g. it may not be possible to sell that many items required given by the solution (222), you could try instead to change the item price if you know in advance the quantity you are able to sell (100). In this case the GoalSeek dialog box values are the same, except now the variable cell is that for the price (C4).

Goal Seek Example #2: Reaching a profit goal, here the intention is to figure out how much sales to make in Q4 in order to achieve some intended target net profit for the year, for e.g. 100,000

Go to Data -> What if Analysis -> Goal Seek.

In the Goal Seek dialog box, define the cells/values to test

- Set cell - the reference to the formula cell that calculates the total net profit (E18).
- To value - the formula result you are trying to achieve (100,000).
- By changing cell - the reference for the input cell that you want to adjust (Gross for Q4 – C17).

And click Ok.

2 Linear Programming Basics: Burger Production

File to use: SumProduct-Demo

This demonstrates the use of the SumProduct function which is useful for computing results from linear functions for the objective / goal function or the linear equations for the constraints.

File to use: Burgers Production - max profit

3 Plant production allocation

File to use: Production allocation - min cost

Problem description:

There are 4 production plants (Plant A, Plant B, Plant C, Plant D) producing product X that located in different geographical areas. These 4 plants are capable of shipping to 4 different destination countries: France, Germany, Poland and Spain.

The main challenge is to determine which plant should supply how many units of the product to each of these destination countries, taking into account transportation costs.

The max production capacity of the plants are as follows (i.e. the plant cannot produce more than these number of units).

Plant	Max Production Capacity
A	10
B	12
C	15
D	9

We also know in advance (possibly through predictive analytics) the demand for number of units of product X from each of these respective destination countries:

Country	Demand in units
France	15
Germany	6
Poland	14
Spain	11

Through previous historical records, we have determined the cost of transporting 1 unit of product X from each of these 4 plants to each of the 4 destination countries based on the table below

	Plant A	Plant B	Plant C	Plant D
France	25	35	36	60
Germany	55	30	25	25
Poland	40	50	80	90
Spain	30	40	66	75

The goal / objective function is the total cost of transporting all units of product X from all plants to the final destination countries. We wish to minimize this total cost while at the same time ensuring that the demand of the required number of units for each country is met.

The decision variables under our control is the number of units produced at each of the respective Plants A – D to be shipped to the respective 4 destination countries.

The worksheet *Initial* shows the initial setup of the LP model.

The worksheet *Solution* shows the final solution with the formulation of the LP model in Excel Solver taking into account the goal function that is to be minimized and the 2 main constraints:

- The total number of units produced from each plant cannot exceed its max capacity
- The number of units assigned to be shipped to each destination country must meet the projected demand for that country